Aman Sharma

Space Robotics Engineer

Contact:

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About me:

I am a robotics engineer and the thing I enjoy the most is learning new things. Throughout my life I have developed various interests including space, robotics, web design and development etc., and my goal is to learn as much as possible about as many things as I can. Currently I am working on a personal website and teaching myself HTML, CSS, Javascript, and Photoshop.

Social:

http://github.com/asharmarx

http://linkedin.com/in/asharmarx

Experience:

June-Oct 2017

Graduate Research Assistant

Aalto University, Finland

Developed a learning platform for a micro-air vehicle (MAV) with vertical takeoff and landing (DJI M100)

Performed system testing and identification using model predictive control (MPC)

June-Aug 2016

Flight Software Engineer

Bitcraze AB, Sweden

Researched and implemented flight stability methods in nano-quadcopters through the optimization of system dynamics and state-estimator methods

Converted a Matlab library used to estimate the location of positioning-anchors (for autonomous flight) into a Python librar

June-Aug '13 | '14

Process Engineer

Intel, Arizona

Validate microprocessor inspection processes through dat mined via windows scripting

Parse and analyze large quantities of data for various processes with MATLAB and JMP analytical software

Jan-April 2014

Tribology Research Intern

Argonne National Laboratory, Illinois

Investigate critical failure mechanisms of bearing and gears in wind turbine drive-train components through bench top testing

Derive, analyze, explain, and present results and conclusions from the gathered data

Education:

2015-2017

MSc | Space Science and Techonology

Luleå University of Technology

2016-2017

MSc | Space Robotics and Automation

Aalto University

Thesis: System Identification of a Drone Platform (DJI M100)

2010-2015

BSE| Aerospace Engineering (Astronautics)

Arizona State University

GPA: 3.81

Skills:

Platforms	
Software	MATLAB/Simulink | Solidworks | ROS | JMP| Microsoft Office
Coding	Python | NumPy | SciPy | Bash | HTML | CSS | Javascript | LaTeX | AHK | C++ | C
Language	English (Fluent) | Hindi (Fluent) | Spanish (2B) | French (2A) | Finnish (1A)
Courses	Feedback Systems | Sensors and Control | Stochastic Models and Estimation Modelling, Estimation, and Dynamic Systems | Robotics: Manipulation, Decision Making and Learning | Embedded Real-Time Systems | Mechatronics

Projects:

DJI M100	Researched and implemented real-time flight stability methods through the use of Model Predictive Control (MPC) as well as researched feature and object detection algorithms for in-flight detection through stereo vision
Lunabotics	Lead a team as a mechanical engineer in order to design, build, and program a lunar robot capable of mining simulated lunar sand for the NASA Lunabotics competition